

I18N LANGUAGE MANAGEMENT

Версия: 1.0.0

Дата: 2026-06-13

Статус: Active — Canonical Source of Truth

Автор: Koda (NLP-Core-Team)

PURPOSE

Этот документ описывает **управление языками Balloo из технической зоны**.

Цель: Обеспечить централизованное управление языками без ручного изменения кода.

TECHNICAL ZONE CAPABILITIES

What Technical Zone Can Do

Capability	Description	Status
View Languages	See all supported languages	☒ Current
View Coverage	See translation coverage per language	☒ Current
Add Language	Add new language metadata	 Future
Upload Translations	Upload translation files via UI	 Future
Validate Translations	Run automated validation	 Future
Enable/Disable	Toggle language availability	 Future
Change Priority	Modify language priority	 Future
View Audit Log	See change history	 Future

ROLES & PERMISSIONS

Management Roles

Role	Permissions	Members
Admin	Full access	CTO, Tech Lead
i18n-team	Language management	i18n developers
Translator	Update translations	Professional translators
Reviewer	Review translations	Native speakers

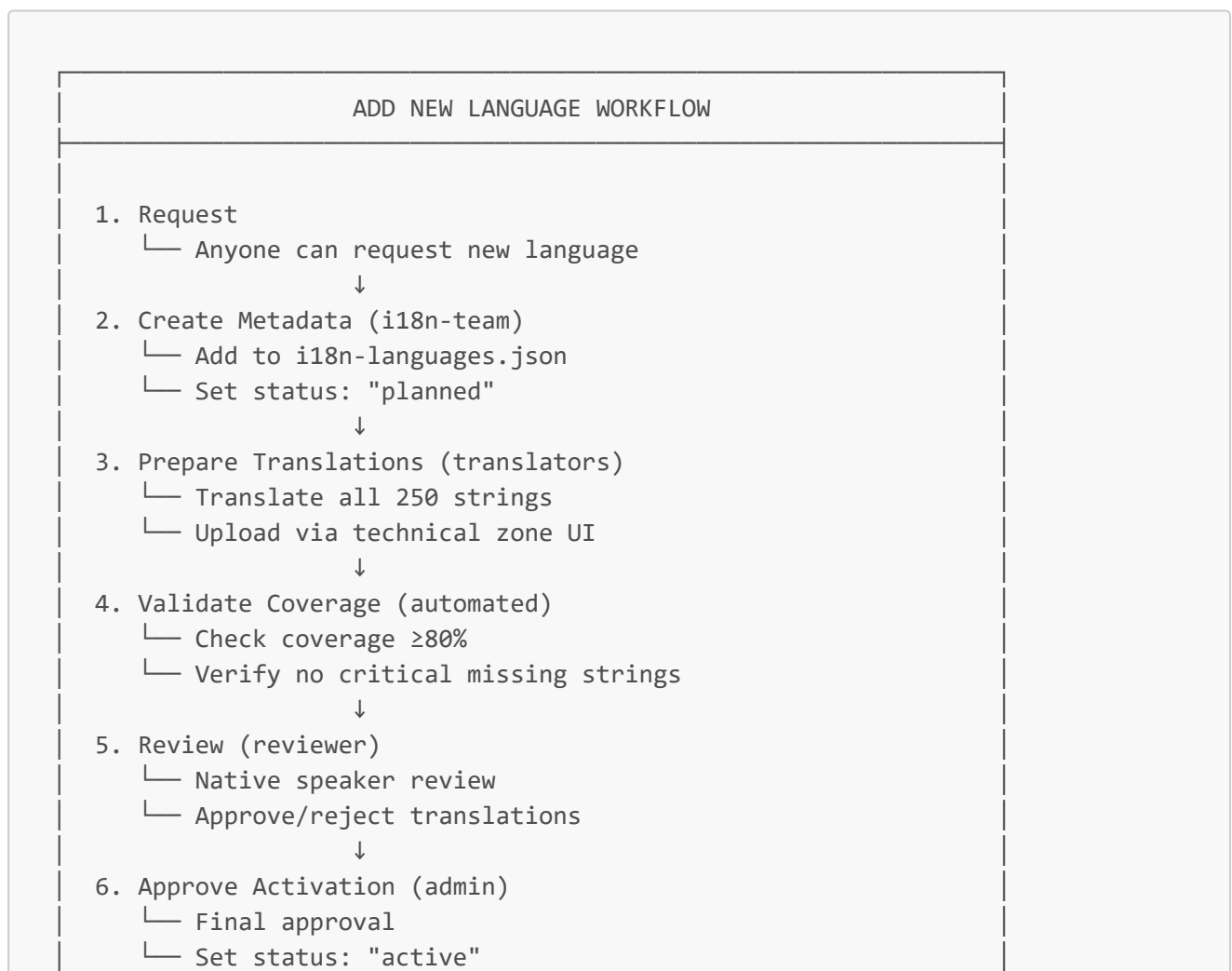
Role	Permissions	Members
User	Select language	End users

Permission Matrix

Action	Admin	i18n-team	Translator	Reviewer	User
Add language	✓	✓	✗	✗	✗
Enable/disable	✓	✓	✗	✗	✗
Change priority	✓	✗	✗	✗	✗
Update translations	✓	✓	✓	✗	✗
Review translations	✓	✓	✗	✓	✗
Select language	✓	✓	✓	✓	✓
View audit log	✓	✓	✗	✗	✗

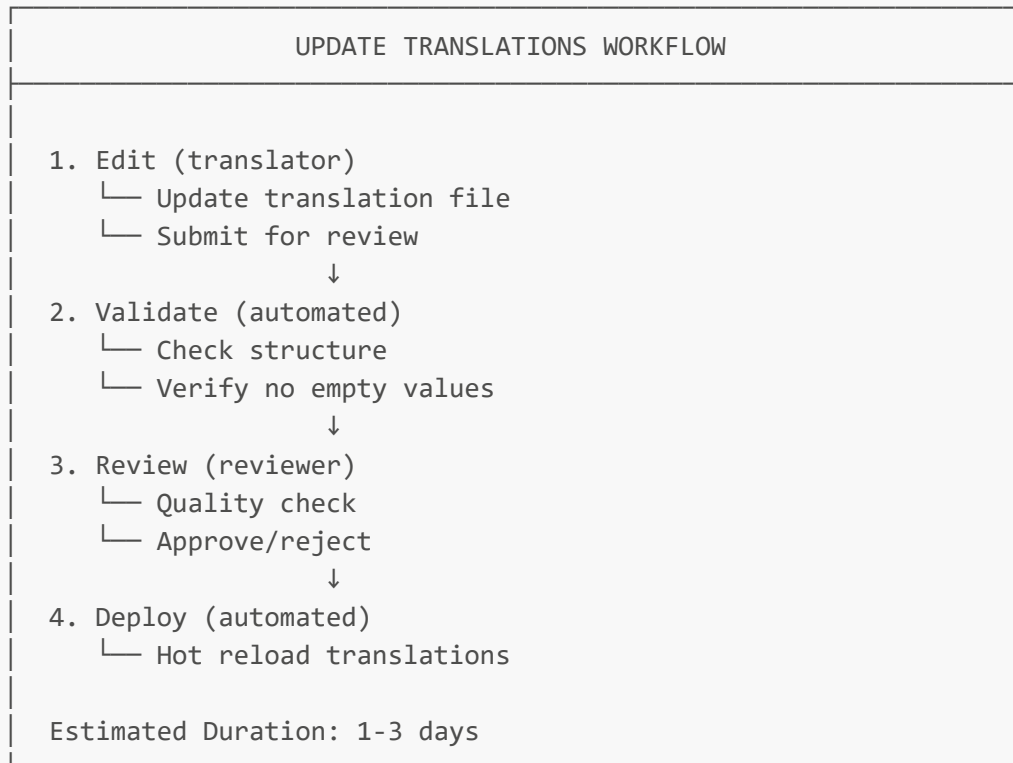
MANAGEMENT WORKFLOWS

Workflow: Add New Language



- ↓
7. Deploy (automated)
 - └ Hot reload or rebuild
 - └ Language available in UI
- Estimated Duration: 7-12 days

Workflow: Update Translations



Workflow: Enable/Disable Language



- └─ Approve/reject
 - ↓
- 4. Update State (i18n-team)
 - └─ Set enabled: true/false
 - └─ Log audit entry
 - ↓
- 5. Deploy (automated)
 - └─ Hot reload

Estimated Duration: 1-2 days

STATE FILES

Source of Truth

File	Purpose	Update Method
<code>i18n-languages.json</code>	Language registry	Technical zone UI
<code>i18n-packages.json</code>	Translation packages	Technical zone UI
<code>i18n-translation-coverage.json</code>	Coverage tracking	Automated
<code>i18n-runtime-map.json</code>	Runtime state	Automated
<code>i18n-management-map.json</code>	Roles & permissions	Admin
<code>i18n-missing-strings.json</code>	Missing strings	Automated

Example: Update Language State

```
// Before (planned)
{
  "languageId": "kk",
  "status": "planned",
  "coverage": 0
}

// After (active)
{
  "languageId": "kk",
  "status": "active",
  "coverage": 100,
  "enabled": true,
  "runtimeAvailability": true,
  "uiAvailability": true
}
```

AUDIT TRAIL

What Is Logged

Event	Details Logged
Language added	Who, when, metadata
Language enabled/disabled	Who, when, reason
Translations updated	Who, when, coverage change
Priority changed	Who, when, old/new priority
Review completed	Who, when, decision

Audit Log Structure

```
{
  "id": "audit-001",
  "timestamp": "2026-06-13T10:00:00Z",
  "actor": "i18n-team",
  "action": "language.add",
  "target": "language:kk",
  "changes": {
    "status": { "from": null, "to": "planned" }
  },
  "status": "approved",
  "reviewer": "admin"
}
```

FUTURE UI FEATURES

Language Management Dashboard

LANGUAGE MANAGEMENT DASHBOARD

Languages (12)

Code	Name	Status	Coverage		Actions
ru	Russian	active	100%		[Edit]
en	English	active	100%		[Edit]
hi	Hindi	active	80%		[Edit]
...					
[+ Add Language]					

Coverage Overview

Average: 96.7%

Missing: 100 strings

Languages below threshold: hi, zh

Recent Activity

[2026-06-13] hi translations updated by translator-1

[2026-06-12] zh reviewed by reviewer-zh

Add Language Form

ADD NEW LANGUAGE

Language ID: [kk _____] (ISO 639 code)

Canonical Name: [Kazakh _____] (English name)

Native Name: [Қазақ _____] (Name in native script)

Category: [Regional _____▼]

Region: [Kazakhstan _____] (For regional languages)

Priority: [13 _____] (Display order)

Translation File

[Upload File] or [Create from Template]

[Cancel] [Save as Planned] [Validate & Activate]



For AI Code Generation

```
{
  "languageManagement": {
    "technicalZone": {
      "capabilities": [
        "view-languages",
        "view-coverage",
        "add-language",
        "upload-translations",
        "validate",
        "enable-disable",
        "change-priority",
        "view-audit"
      ],
      "roles": ["admin", "i18n-team", "translator", "reviewer", "user"],
      "workflows": ["add-language", "update-translations", "enable-disable"],
      "stateFiles": [
        "i18n-languages.json",
        "i18n-packages.json",
        "i18n-translation-coverage.json",
        "i18n-management-map.json"
      ]
    }
  }
}
```

RELATED DOCUMENTS

- [I18N_POLICY.md](#) — Language policy
- [I18N_MANAGEMENT_MODEL.md](#) — Management model
- [I18N_NEW_LANGUAGE_PLAYBOOK.md](#) — Add language guide
- [../state/i18n-management-map.json](#) — Management mapping

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 **Baloo** - Проверни общение!